

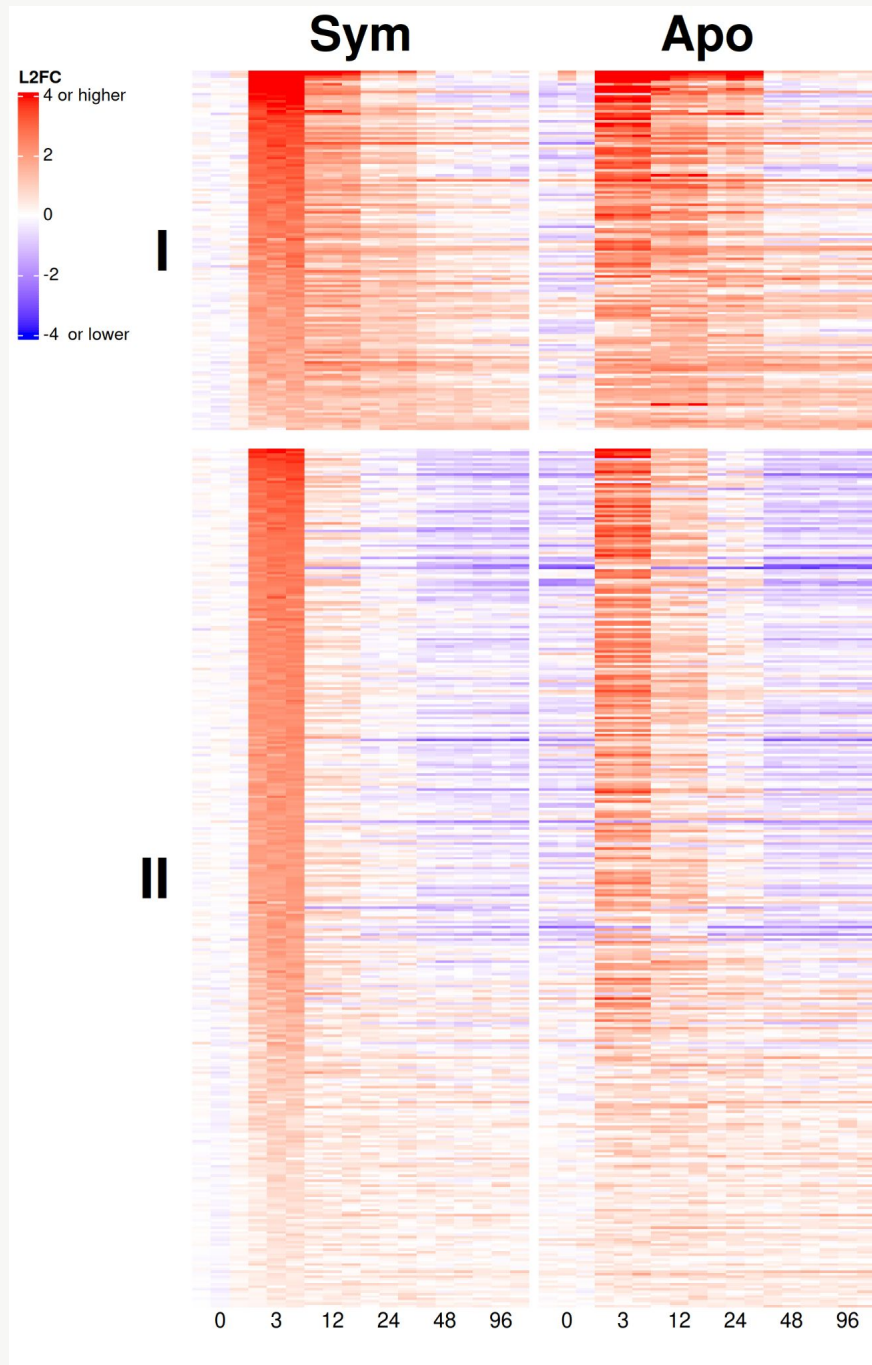
Independent Reproduction of Cleves et al. (2020)

*Insights into coral bleaching under heat stress
from analysis of gene expression in a sea
anemone model system*

All figures in this presentation are recreations of figures from Cleves et al. (2020), generated from my independent reanalysis of the published data. Figure descriptions are adapted from the corresponding descriptions in the original paper.

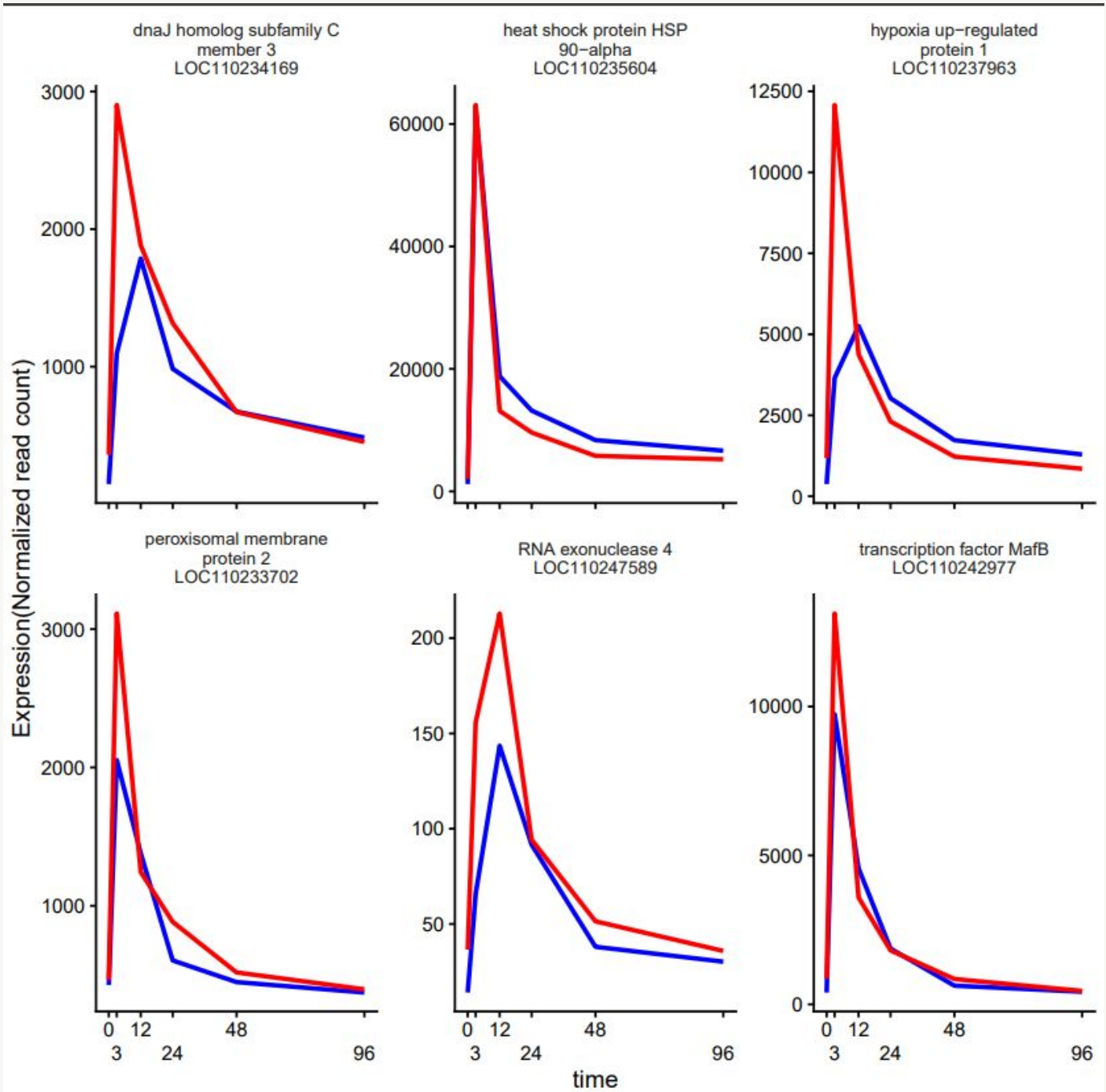
Cleves PA, Krediet CJ, Lehnert EM, Onishi M, Pringle JR. Insights into coral bleaching under heat stress from analysis of gene expression in a sea anemone model system. *Proc Natl Acad Sci U S A*. 2020;117(46):28906-28917.
doi:10.1073/pnas.2015737117

Figure 2A



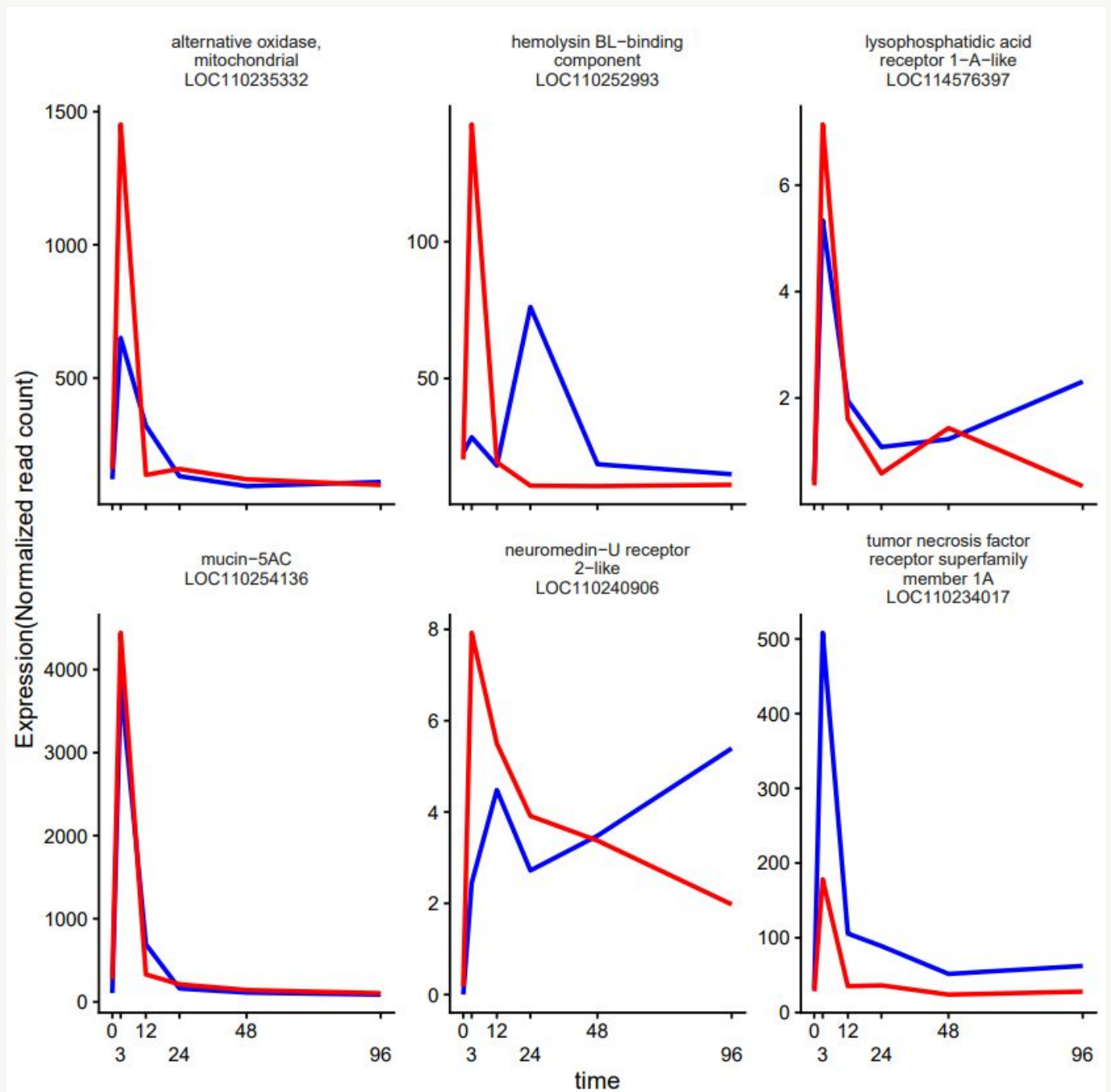
Heatmaps of genes significantly and highly up-regulated in symbiotic anemones in response to a shift from 27 °C to 34 °C (Left) and of the same genes in aposymbiotic anemones (Right). Genes were grouped into two clusters by k-means clustering based on their expression patterns in the symbiotic animals and are sorted by their extents of up-regulation from 0 to 3 h in those animals. Expression values for each gene in the aposymbiotic animals (Right column) were also normalized to the value for that gene in the symbiotic animals at time = 0, and the genes are shown in the same order as in the Left column. Colors in the heatmap are based on a log2 scale (as shown).

Figure 2B - Cluster I



Expression patterns for example genes in cluster I. Data are shown as raw read counts normalized to library size. Red, data for symbiotic animals; blue, data for aposymbiotic animals.

Figure 2B - Cluster II



Expression patterns for example genes in cluster II. Data are shown as raw read counts normalized to library size. Red, data for symbiotic animals; blue, data for aposymbiotic animals.

Figure 2C - Cluster I

GO.ID	Term	Annotated	Significant	Expected	weightFisher
GO:0042026	protein refolding	12	6	0.07	3.2e-11
GO:0034620	cellular response to unfolded protein	35	8	0.21	6.4e-10
GO:0006457	protein folding	90	16	0.54	2.1e-09
GO:0034976	response to endoplasmic reticulum stress	45	6	0.27	0.00016
GO:0030968	endoplasmic reticulum unfolded protein r...	26	3	0.16	0.00048
GO:0043123	positive regulation of canonical NF-kapp...	8	2	0.05	0.00096
GO:0050821	protein stabilization	9	2	0.05	0.00123
GO:0034975	protein folding in endoplasmic reticulum	10	2	0.06	0.00153
GO:0046330	positive regulation of JNK cascade	50	3	0.30	0.00325
GO:0035973	aggrephagy	1	1	0.01	0.00596
GO:1903298	negative regulation of hypoxia-induced i...	1	1	0.01	0.00596
GO:0071528	tRNA re-export from nucleus	1	1	0.01	0.00596
GO:1990575	mitochondrial L-ornithine transmembrane ...	2	1	0.01	0.01189
GO:0034605	cellular response to heat	3	1	0.02	0.01778
GO:0010923	negative regulation of phosphatase activ...	3	1	0.02	0.01778
GO:0030888	regulation of B cell proliferation	3	1	0.02	0.01778
GO:0007266	Rho protein signal transduction	35	2	0.21	0.01836
GO:0007160	cell-matrix adhesion	40	2	0.24	0.02361
GO:0000423	mitophagy	4	1	0.02	0.02364
GO:0016192	vesicle-mediated transport	424	6	2.53	0.03516

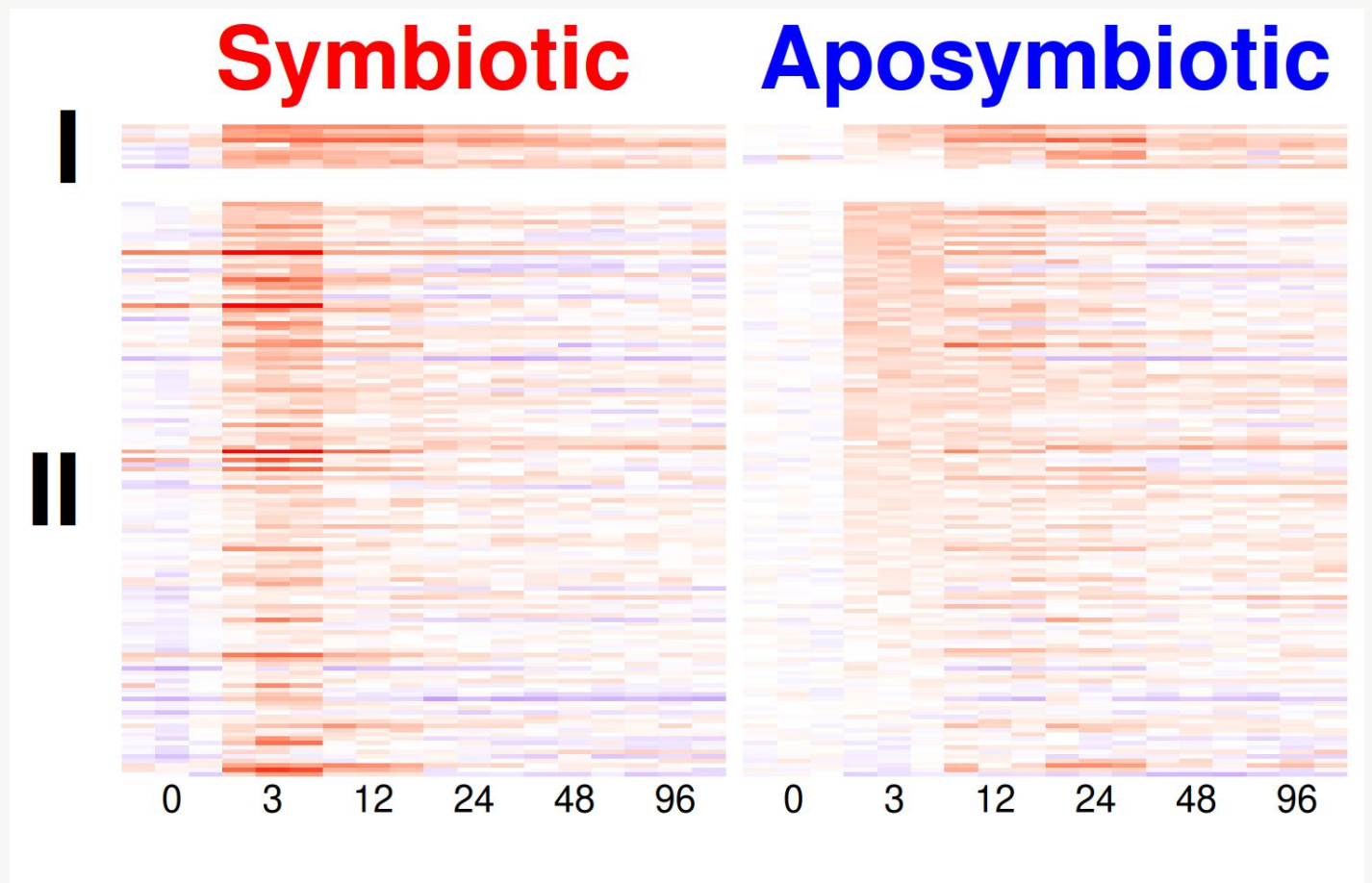
Results from GO-term-enrichment analyses of each cluster.

Figure 2C - Cluster II

GO.ID	Term	Annotated	Significant	Expected	weightFisher
GO:0043122	regulation of canonical NF-kappaB signal...	55	8	0.75	7.3e-07
GO:0042981	regulation of apoptotic process	125	10	1.71	1.3e-05
GO:0033209	tumor necrosis factor-mediated signaling...	41	6	0.56	1.8e-05
GO:0070534	protein K63-linked ubiquitination	43	6	0.59	2.4e-05
GO:0046330	positive regulation of JNK cascade	50	6	0.69	5.8e-05
GO:0008298	intracellular mRNA localization	2	2	0.03	0.00019
GO:2001256	regulation of store-operated calcium ent...	4	2	0.05	0.00110
GO:0031954	positive regulation of protein autophosp...	5	2	0.07	0.00182
GO:0019941	modification-dependent protein catabolic...	291	8	3.99	0.00263
GO:0032481	positive regulation of type I interferon...	6	2	0.08	0.00270
GO:0007165	signal transduction	2537	50	34.77	0.00289
GO:0016567	protein ubiquitination	264	13	3.62	0.00355
GO:0043409	negative regulation of MAPK cascade	8	2	0.11	0.00495
GO:0016239	positive regulation of macroautophagy	8	2	0.11	0.00495
GO:0006357	regulation of transcription by RNA polym...	673	18	9.22	0.00537
GO:0061709	reticulophagy	10	2	0.14	0.00781
GO:0002218	activation of innate immune response	11	2	0.15	0.00946
GO:0008284	positive regulation of cell population p...	33	3	0.45	0.01019
GO:0007266	Rho protein signal transduction	35	3	0.48	0.01198
GO:0034097	response to cytokine	61	7	0.84	0.01324

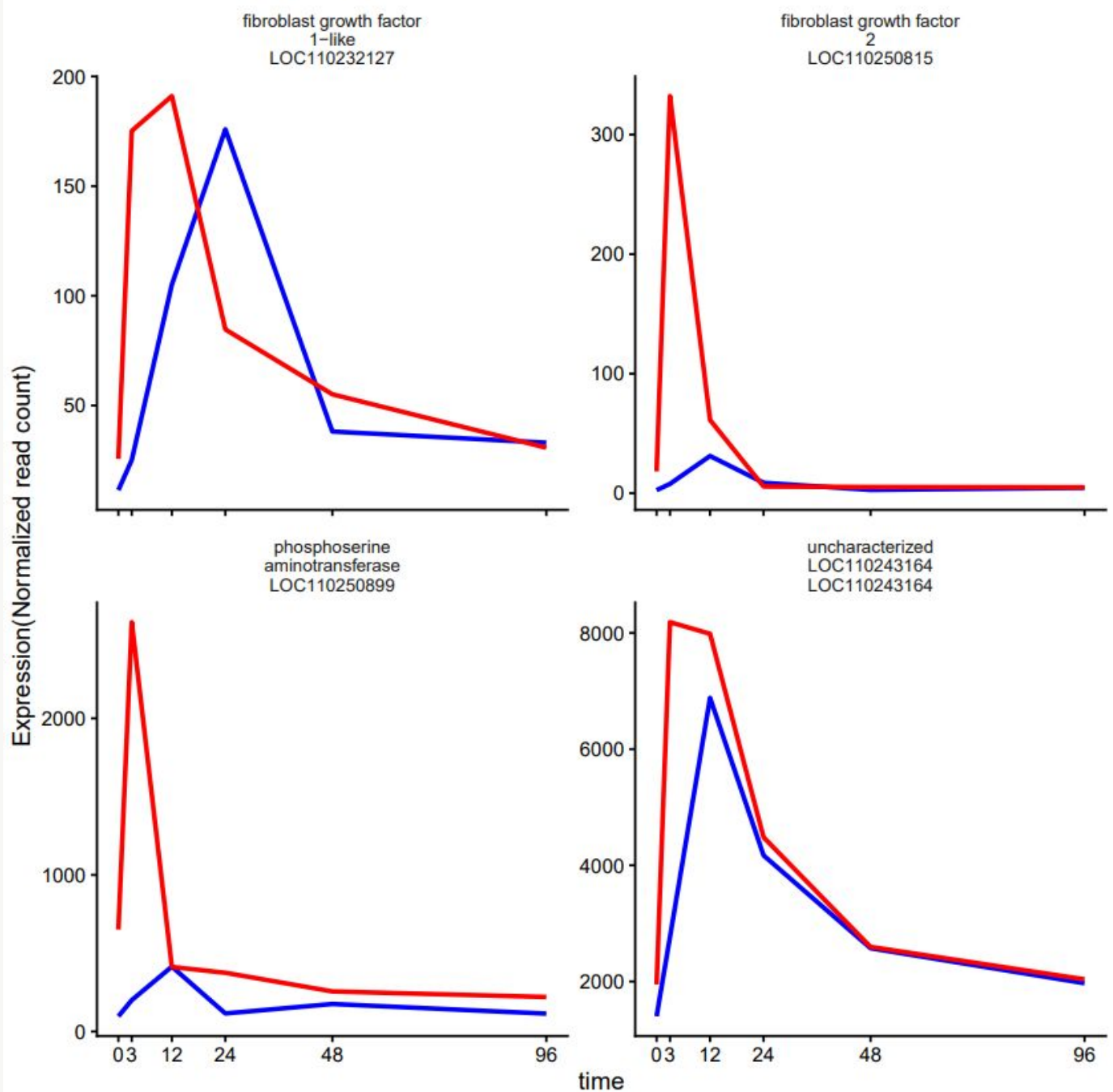
Results from GO-term-enrichment analyses of each cluster.

Figure 2D



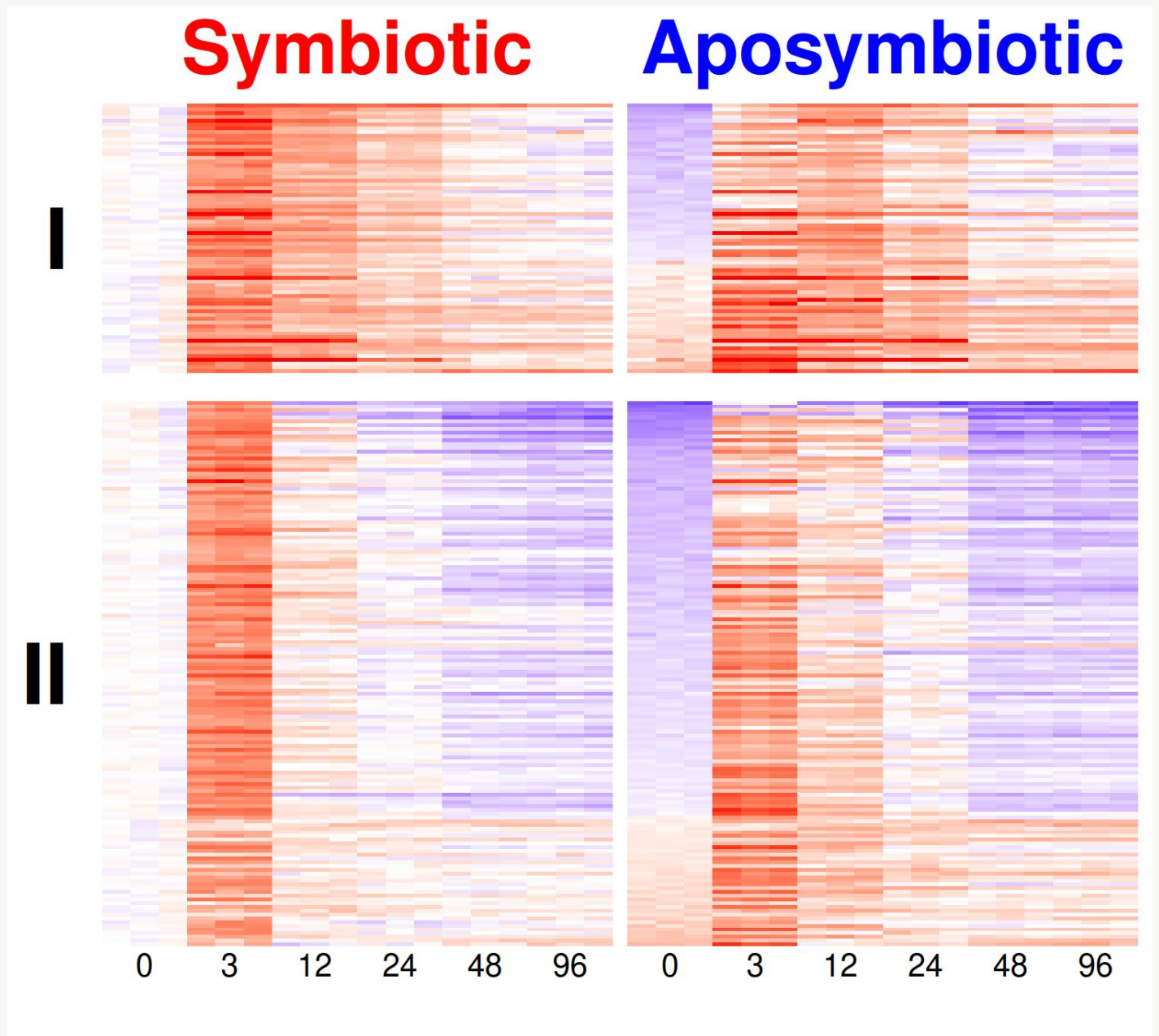
Heatmaps (color scale as in A) showing genes from clusters I and II after normalizing each gene's expression level to its mean value in aposymbiotic animals at time = 0 and then sorting by the degree of up-regulation from 0 to 3 h in aposymbiotic animals. Shown here are only the genes in which up-regulation in the aposymbiotic animals was less than twofold at 3 h

Figure 2E



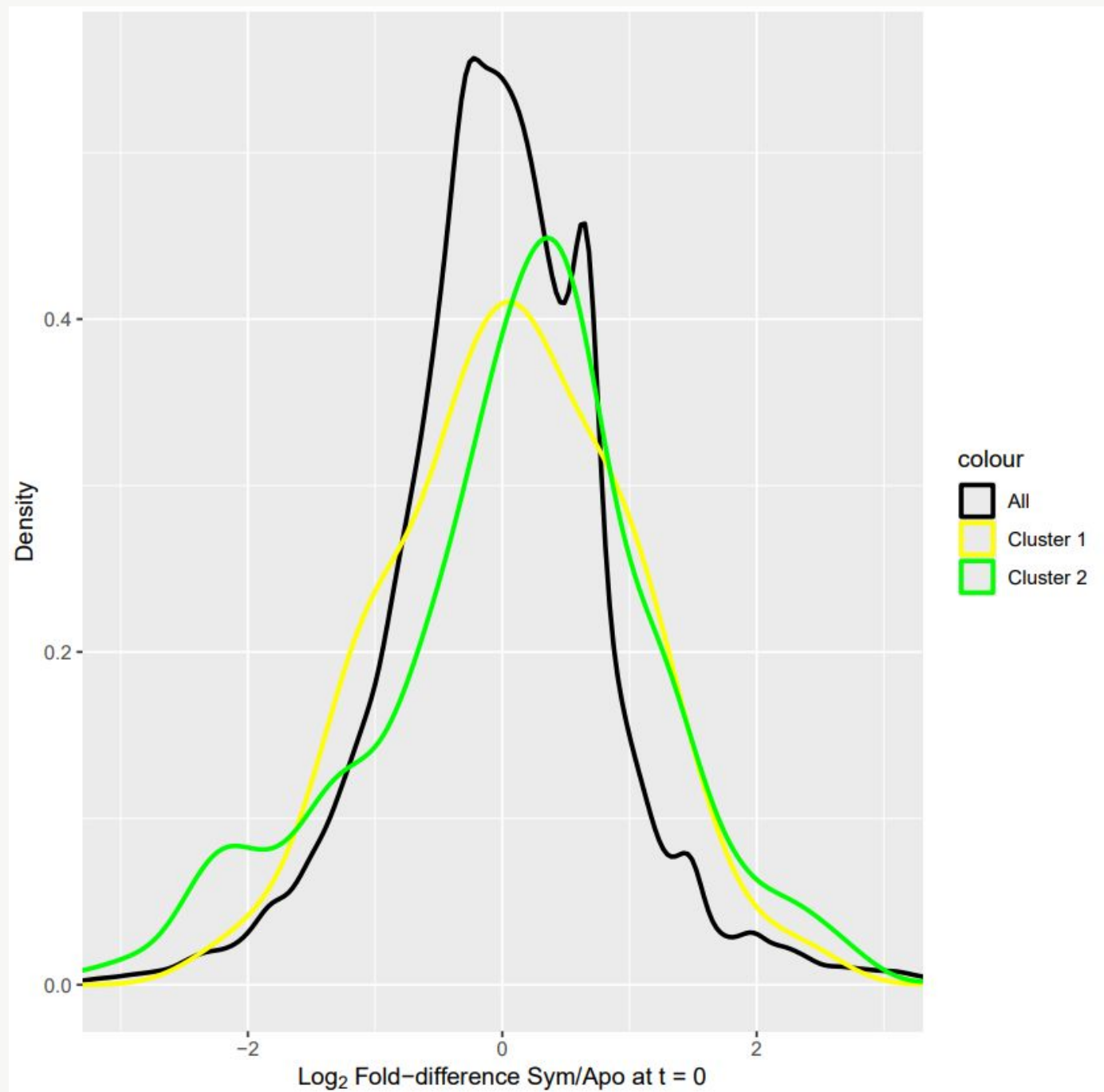
Examples of genes showing differential regulation between symbiotic and aposymbiotic anemones in Cluster I (top row) and II (bottom row). Red, data for symbiotic animals; blue, data for aposymbiotic animals.

Figure 2F



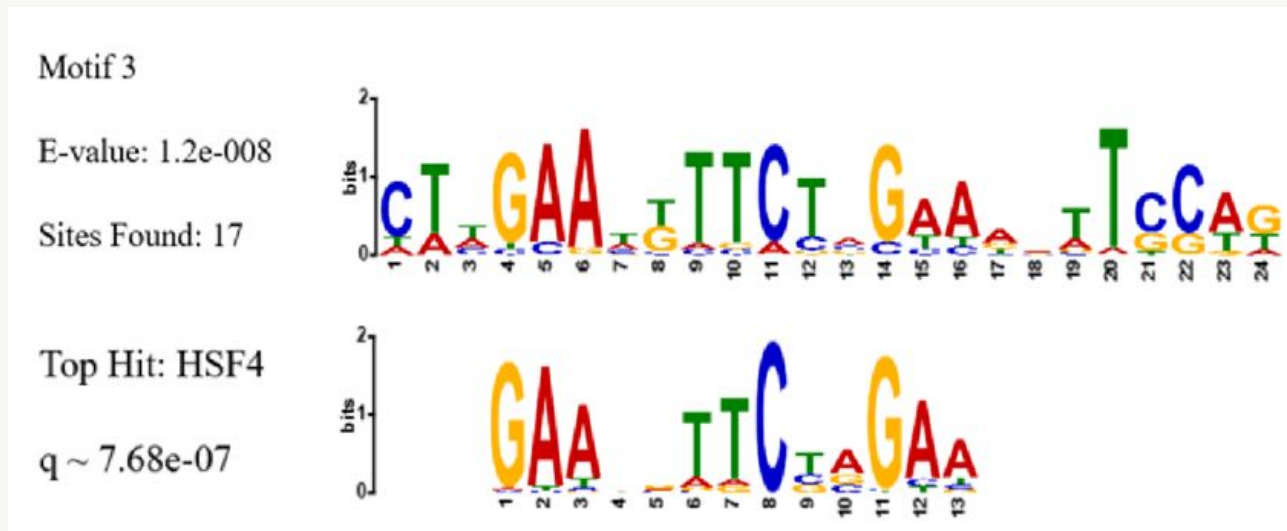
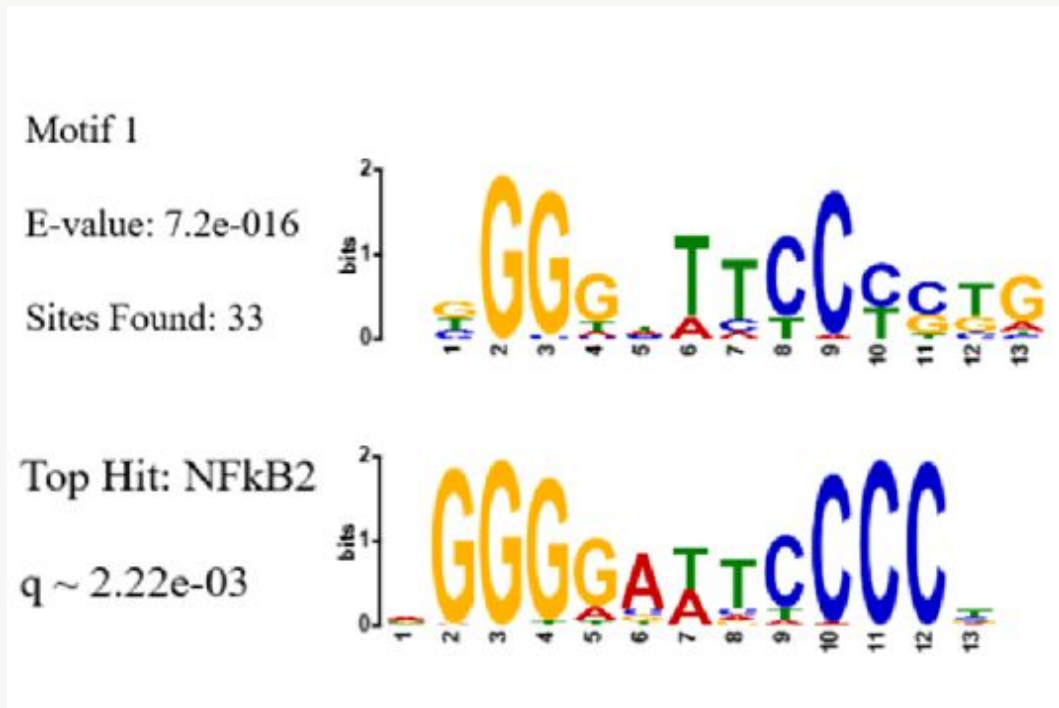
Heatmaps (color scale as in A) of genes as in A but after resorting based on the fold difference between symbiotic and aposymbiotic animals at time = 0. All genes are shown in which this difference was statistically significant (adjusted P value <0.05), independent of the magnitude of the expression difference or of whether expression was higher at 0 h in symbiotic or aposymbiotic animals.

Figure 2G



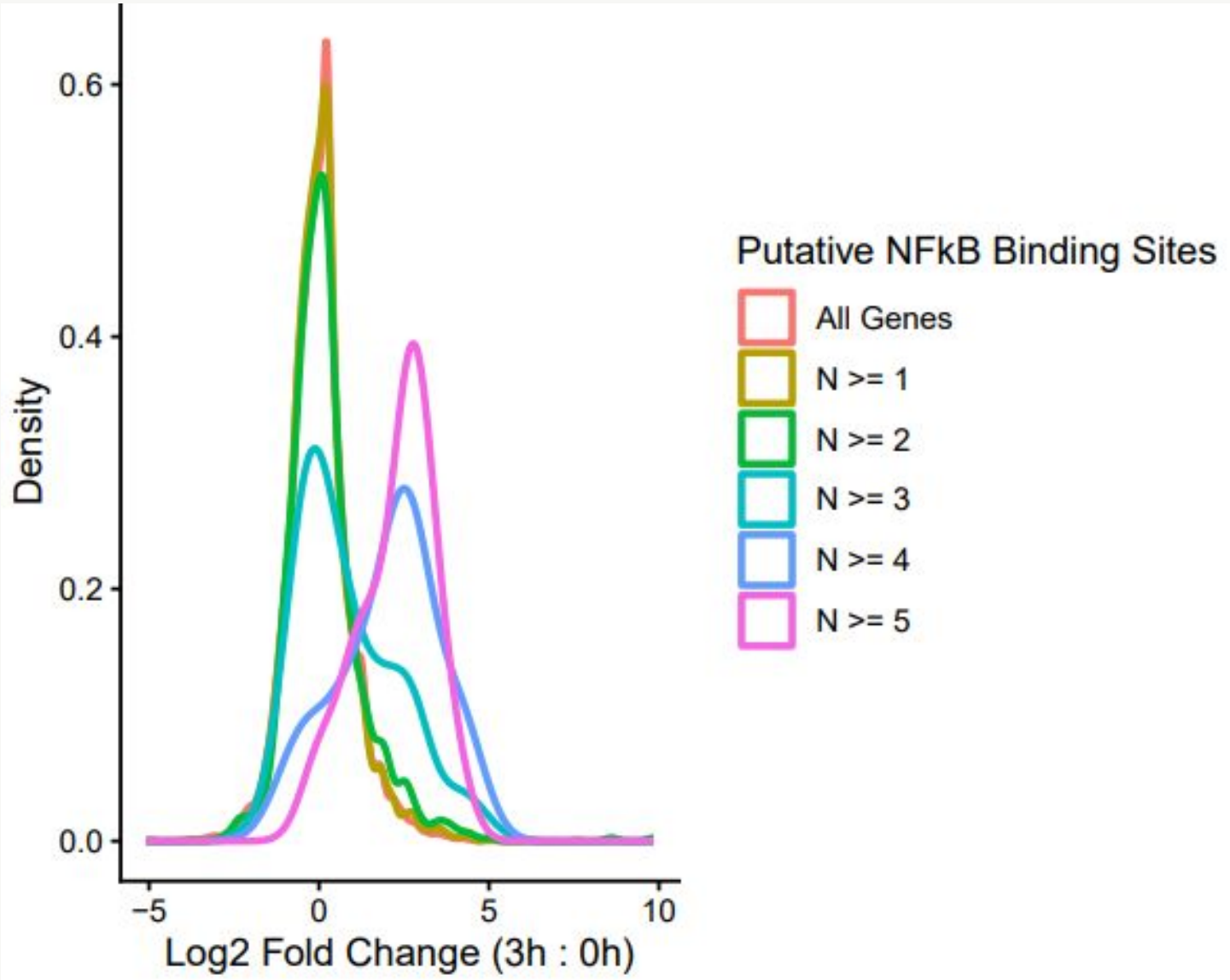
Smoothened-density histograms of log₂ fold differences between symbiotic and aposymbiotic animals at 0 h.

Figures 3B & 3C



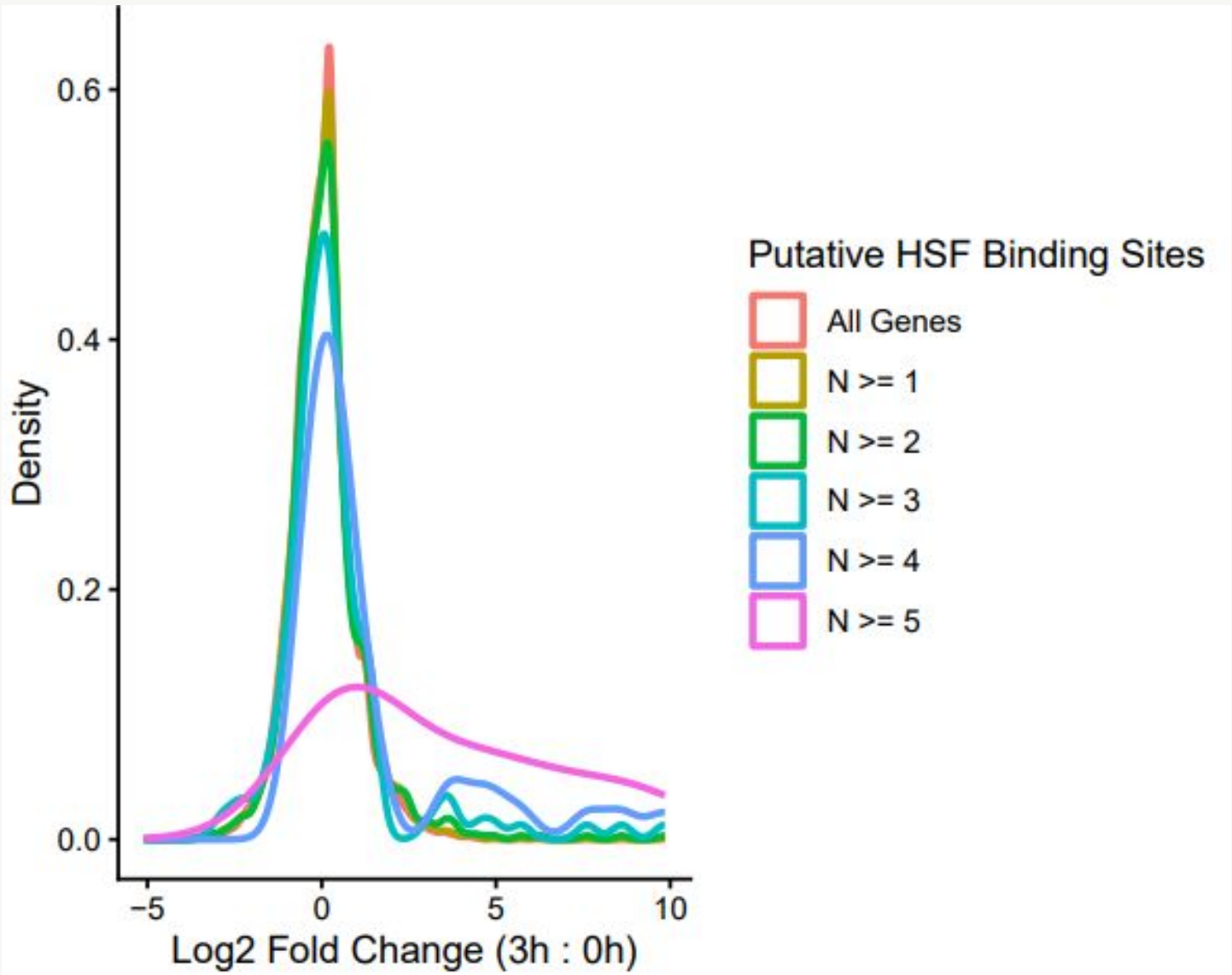
Logo plots showing two of the top four hits returned by the MEME search (Upper plots) and their correspondence to canonical binding sites (Lower plots) for transcription factors NFkB (above) and HSF4 (below)

Figure 3D



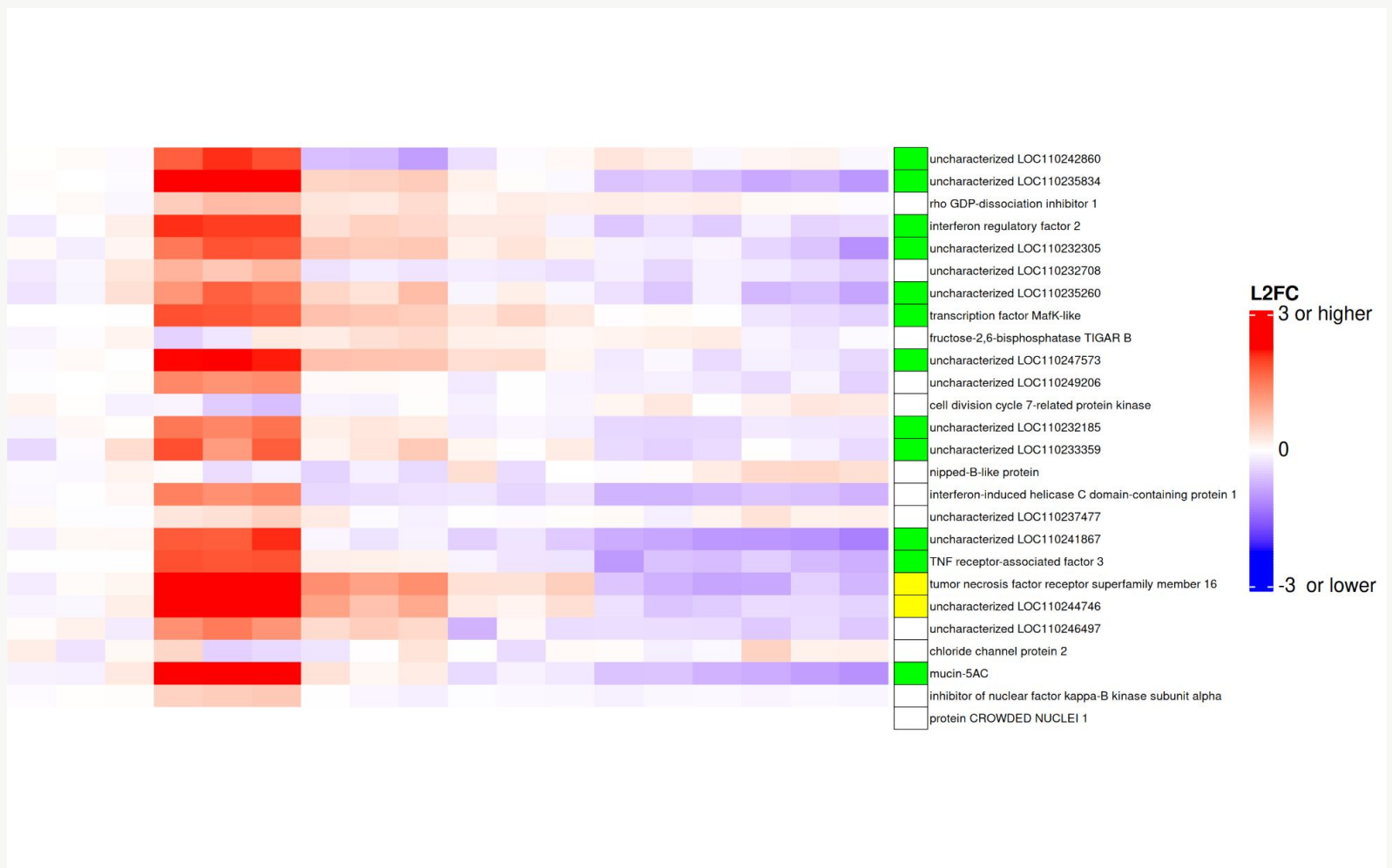
Smoothened-density histogram for the indicated numbers of binding sites plotted as a function of the log2 fold changes in transcript levels between 0 and 3 h after the shift to 34 °C for all genes in the genome.

Figure 3E



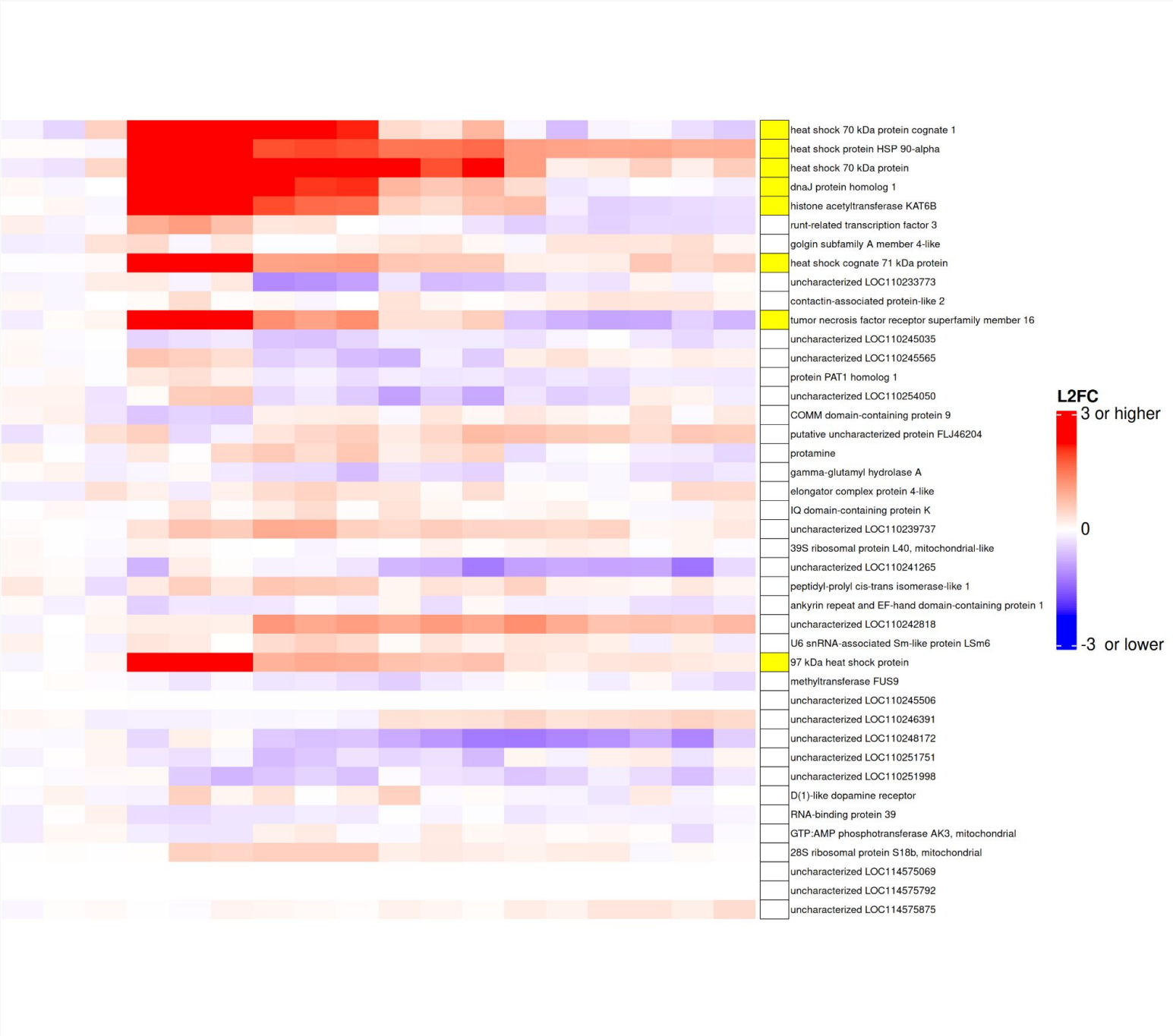
Smoothened-density histogram for the indicated numbers of binding sites plotted as a function of the log2 fold changes in transcript levels between 0 and 3 h after the shift to 34 °C for all genes in the genome.

Figure 3F



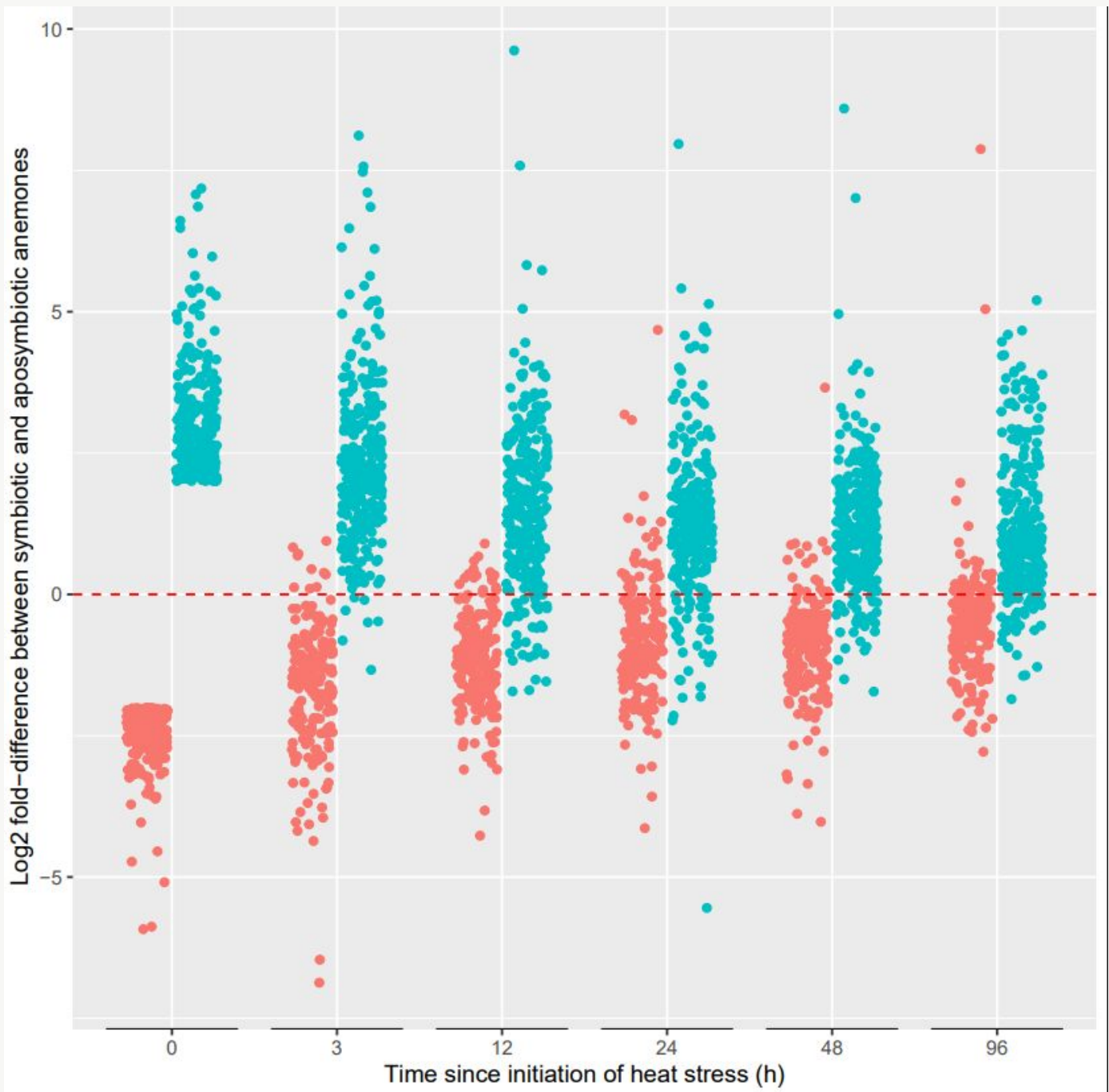
Heatmaps showing changes in transcript levels (log2 scale; see key) as a function of time since the initiation of heat stress for all of the genes with four or more putative NFκB binding sites. Genes marked with yellow are in cluster 1. Those marked with green are in cluster 2.

Figure 3G



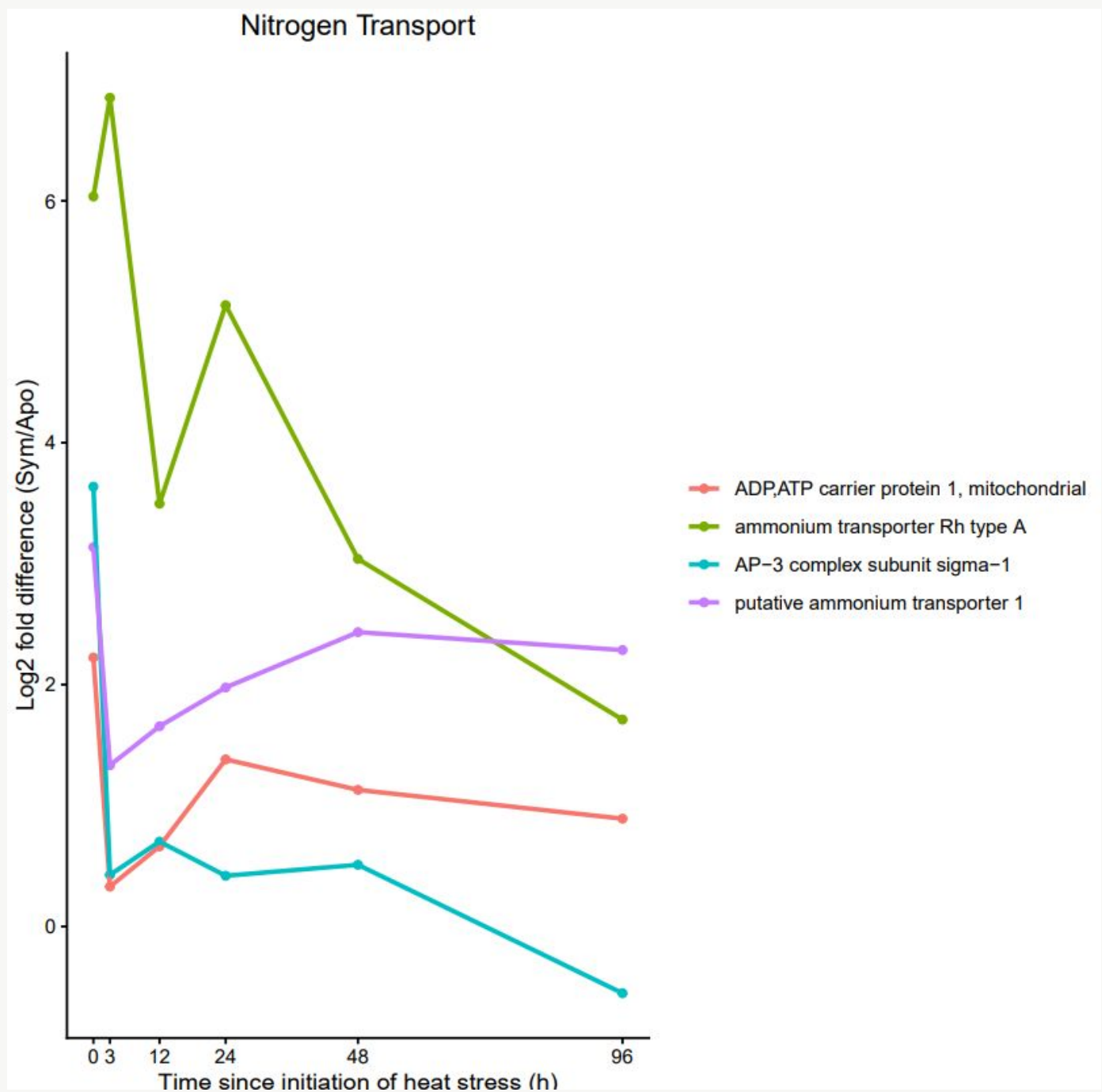
Heatmaps showing changes in transcript levels (log2 scale; see key) as a function of time since the initiation of heat stress for all of the genes with four or more putative HSF binding sites. Genes marked with yellow are in cluster 1. Those marked with green are in cluster 2.

Figure 4A



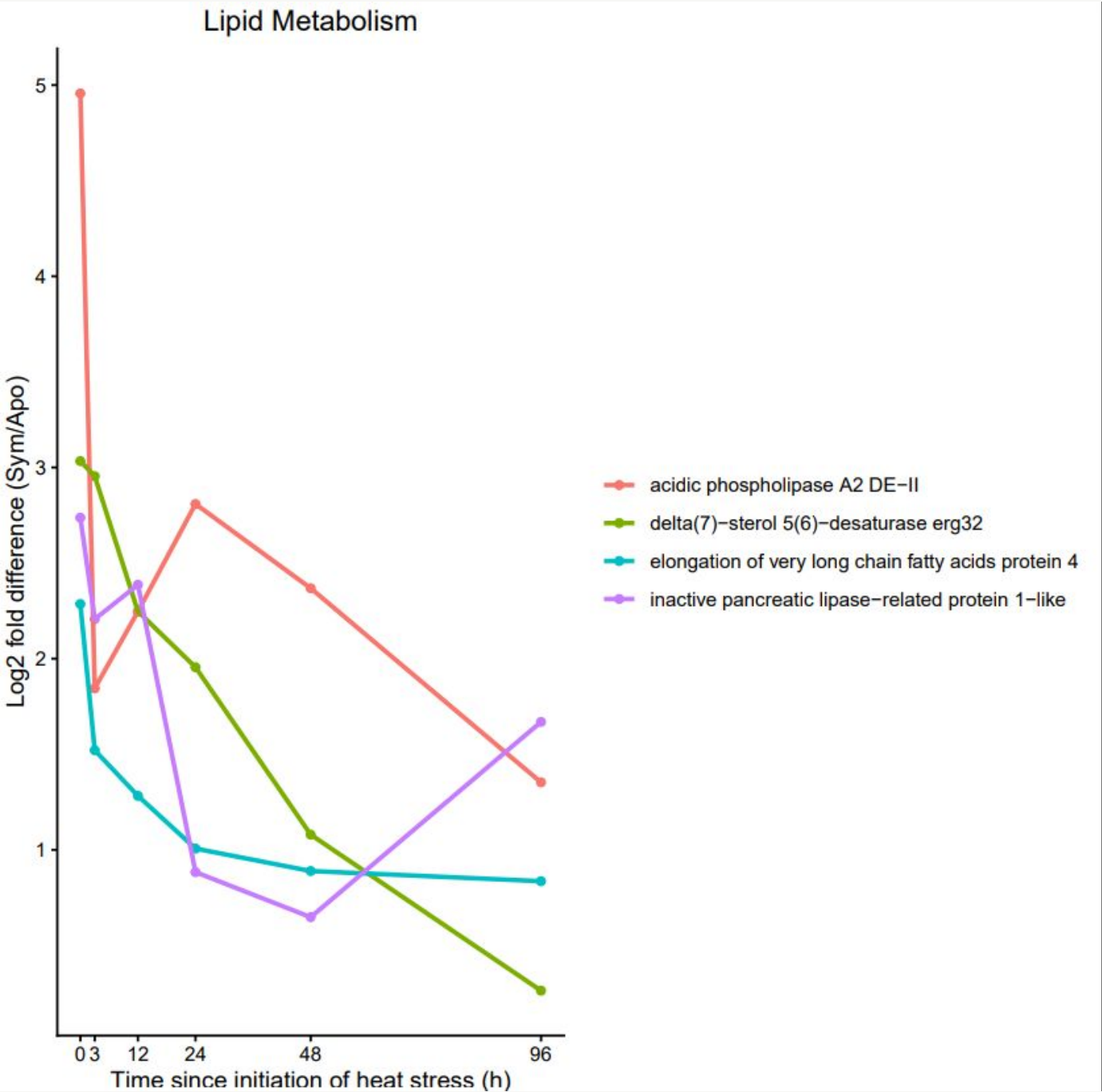
Dot plots of the ratios of expression levels between symbiotic and aposymbiotic anemones for individual genes (log2 scale) as a function of time after the temperature shift. Both genes expressed at higher levels at time = 0 in symbiotic anemones (teal) and those expressed at higher levels in aposymbiotic anemones (salmon) are shown.

Figure 4B



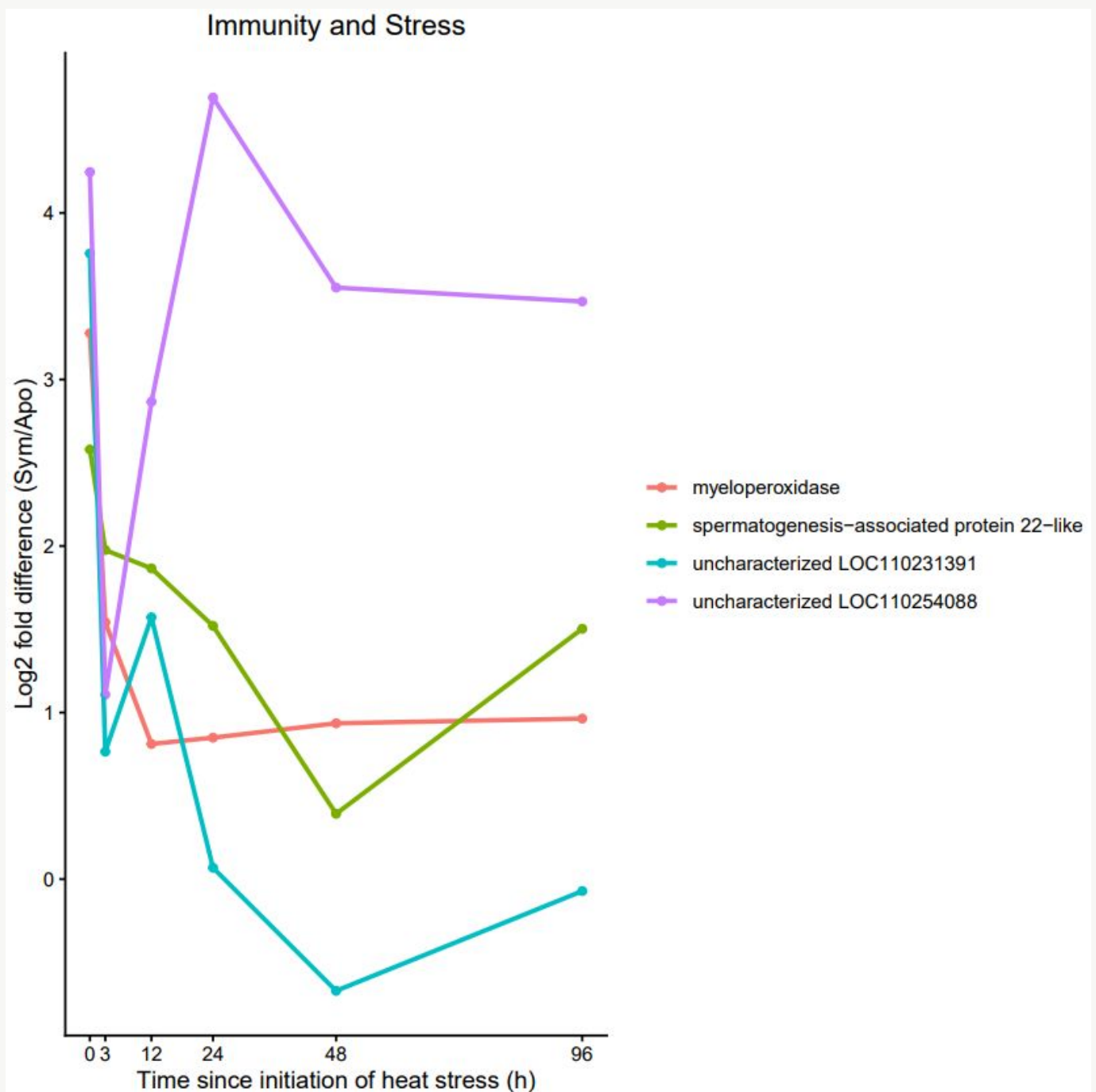
Examples of genes whose expression is rapidly down-regulated in symbiotic anemones after the temperature shift. The mean expression values for each gene at each time in symbiotic anemones are shown relative to the mean expression of that gene in aposymbiotic anemones

Figure 4D



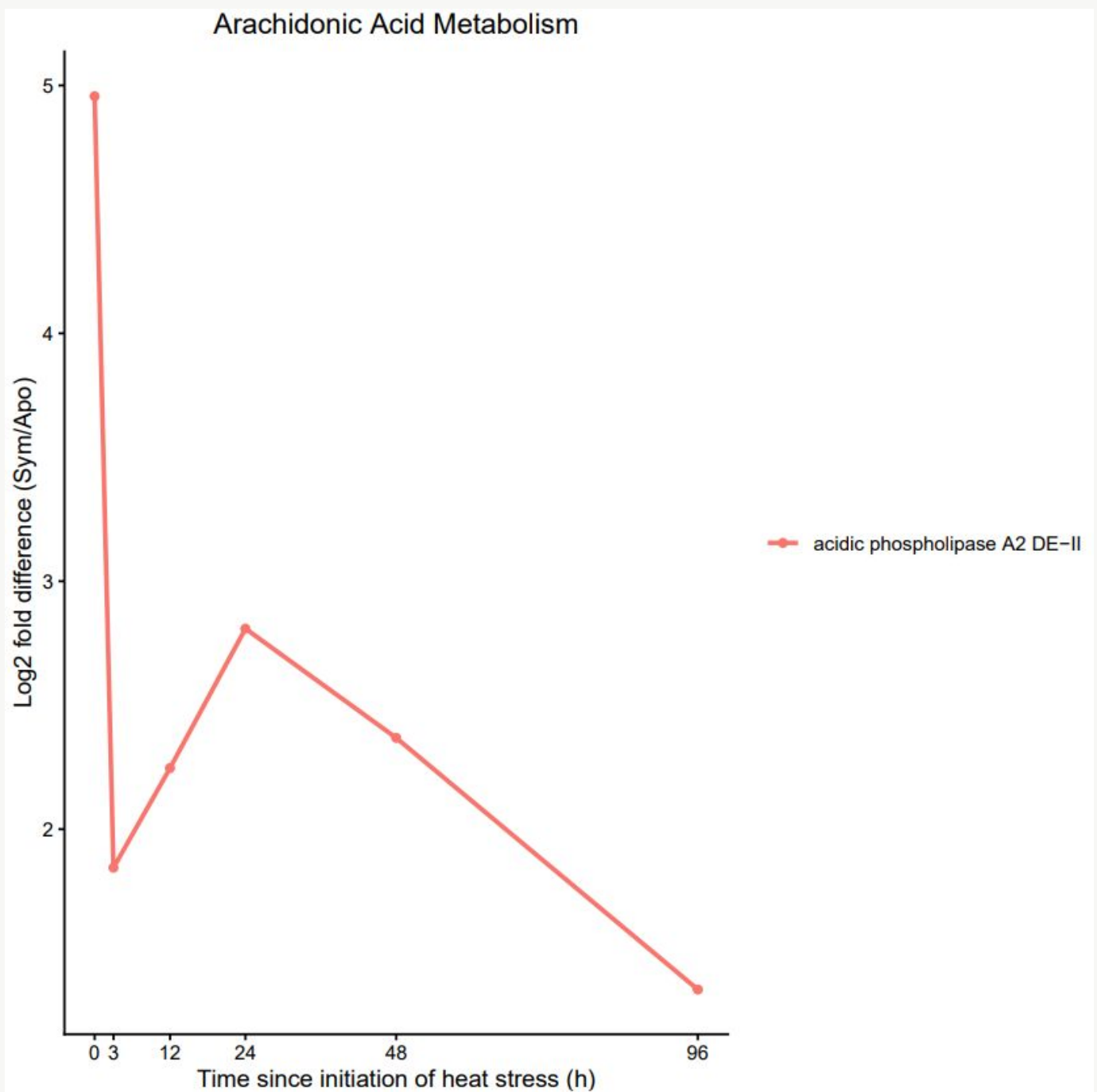
Examples of genes whose expression is rapidly down-regulated in symbiotic anemones after the temperature shift. The mean expression values for each gene at each time in symbiotic anemones are shown relative to the mean expression of that gene in aposymbiotic anemones

Figure 4F



Examples of genes whose expression is rapidly down-regulated in symbiotic anemones after the temperature shift. The mean expression values for each gene at each time in symbiotic anemones are shown relative to the mean expression of that gene in aposymbiotic anemones

Figure 4G



Examples of genes whose expression is rapidly down-regulated in symbiotic anemones after the temperature shift. The mean expression values for each gene at each time in symbiotic anemones are shown relative to the mean expression of that gene in aposymbiotic anemones